



Syntax V

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Please pick up your **name card** from the front

office hours: Mondays, 2:00-3:00 pm

Basement, Room 003, **Linguistics Building**

Assignment II

that

that person (determiner, D)

I know that you are right. (complementiser, C)

the spy saw **the agent** **with** **the binoculars** on the roof

the spy saw **the agent** **with** **the binoculars** on the roof

START REVIEWING for the Midterm

Oct 14, class time in the same classroom

you are allowed to have a cheat sheet
but you have to make your own

You may bring to the exam **one sheet of paper with any (printed or handwritten) notes you choose** -- 8.5 × 11 in. (standard US letter paper size) or smaller, front and back -- that you may consult during the exam, but otherwise it is a closed-book, closed-notes exam

Previously on 201

Phrase Structure Rules (PSRs) for VP, TP

- **VP** $\rightarrow$ (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)
- **TP** $\rightarrow$ NP (**T**) VP
- *Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.*
- steps of drawing **trees** for simple sentences

Steps for Drawing Trees

1. write the sentence at the bottom of the page – leave yourself **plenty of room**
2. label the **part of speech** of each word
3. sentences have a subject **NP** and predicate **VP** – try to identify these first, and draw the subject
4. trees should ALWAYS follow **PSRs**, so try to identify elements that can be part of rules for different phrases
5. every **noun**, **adjective**, **adverb**, **preposition** has its own corresponding phrase **NP**, **AdjP**, **AdvP**, **PP**
6. smaller trees first, then figure out how they connect to the rest of the tree

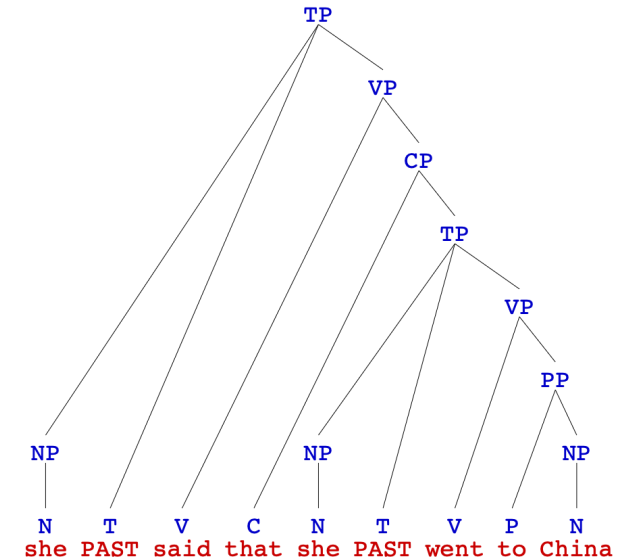
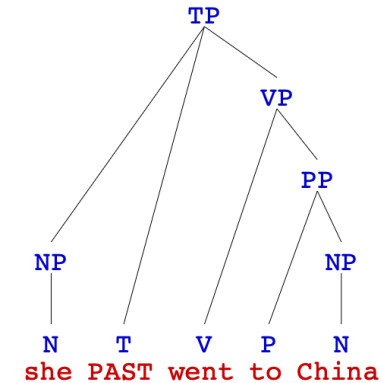
some notes

- VPs can have lots of elements, so figure them out last
- **pay attention to what modifies what** – any phrase that modifies a word should be its **sibling**
- because of the way English is structured, it can often be easier to start at the end and work **backwards**

John said Mary went to the store quickly.

Simple and Complex Sentences

- so far we've been studying **simple sentences**:
sentences with **one** subject **NP** and one predicate **VP**
- now we're going to talk about **complex sentences**: sentences that have another sentence (clause) **embedded** inside of them
 - She said [that **she went to China**].*
 - I bought an umbrella [because **it was raining**].*
 - That's the movie [**I saw**].*
 - The book [that **I read**] was very interesting.*



Complementiser Phrases (CP)

consider the following sentence:

- *Tina thinks that her brother has finished his homework.*

is the underlined string a constituent?

Complementiser Phrases (CP)

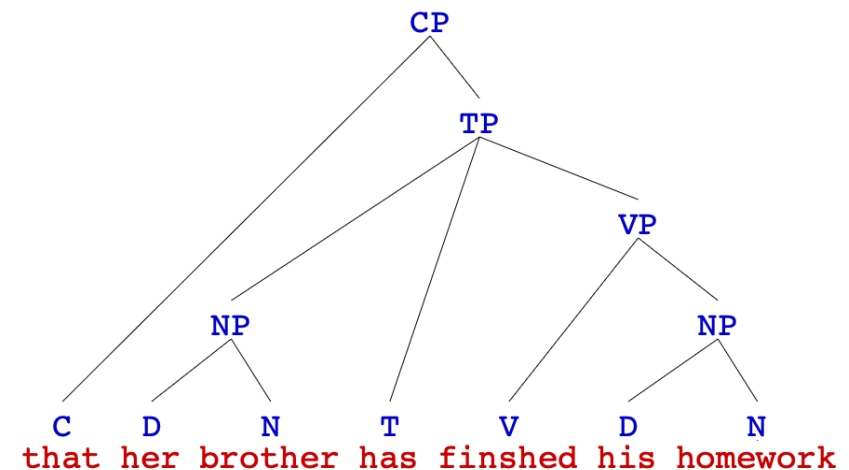
consider the following sentence:

- Tina thinks that her brother has finished his homework.

is the underlined string a constituent?

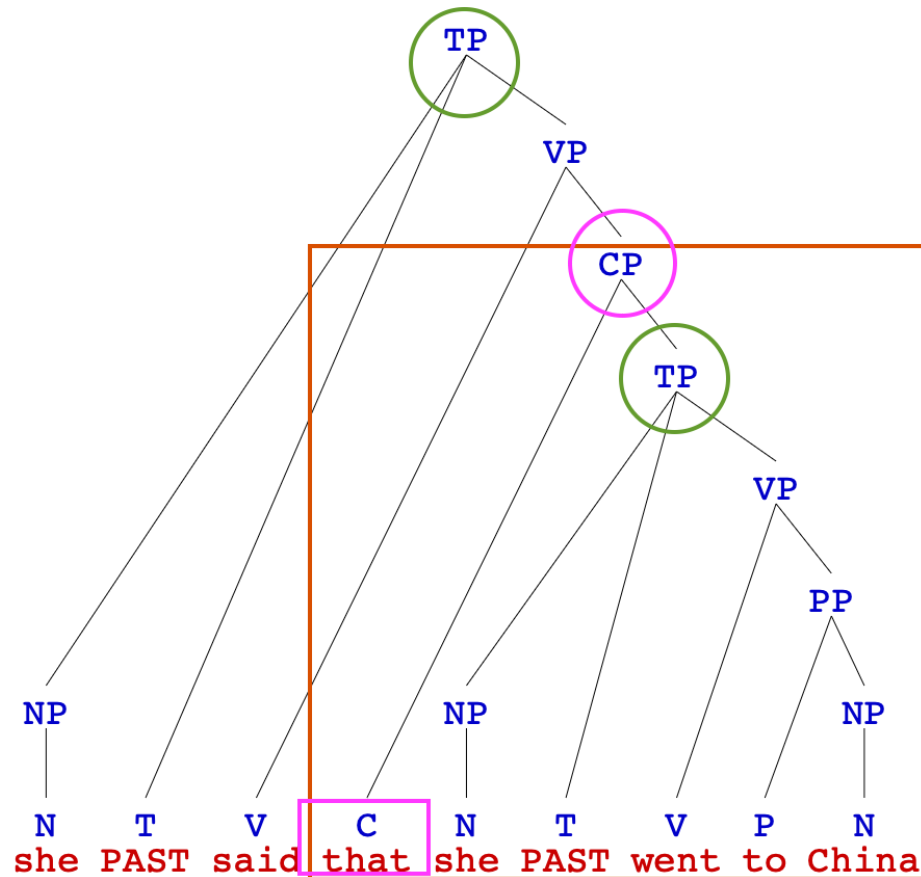
- **movement test**
That her brother has finished his homework
is what Tina thinks.
- **fragment test**
What does Tina think?
That her brother has finished his homework.

CPs are sentences preceded by a **complementiser**



'that her brother has finished his homework'

Complex Sentences



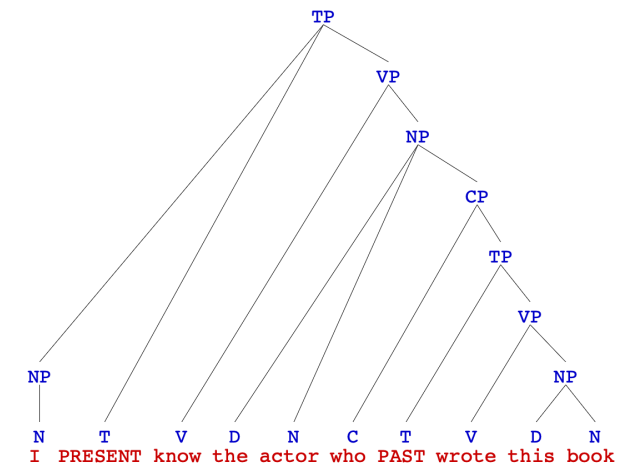
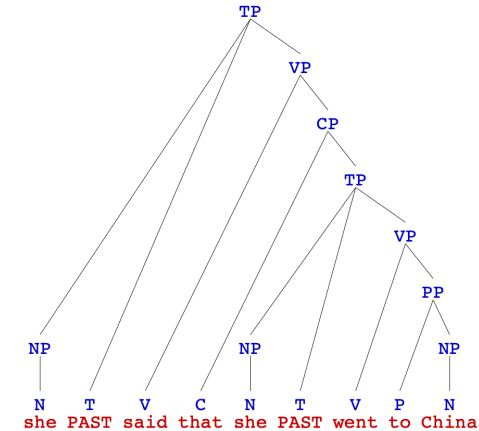
- A **complementiser** can connect two **TPs** together
- **two parts** of a complex sentence:
 - **dependent clause** (aka subordinate clause, embedded clause, embedded sentence) – the sentence that has been **inserted** into the larger sentence
 - *She said [that **she went to China**].*
 - **main clause** (aka independent clause, main sentence) – the **full** sentence, technically **including the dependent clause**, since the dependent clause is inside of the main clause
 - ***She said [that she went to China].***

Complementisers

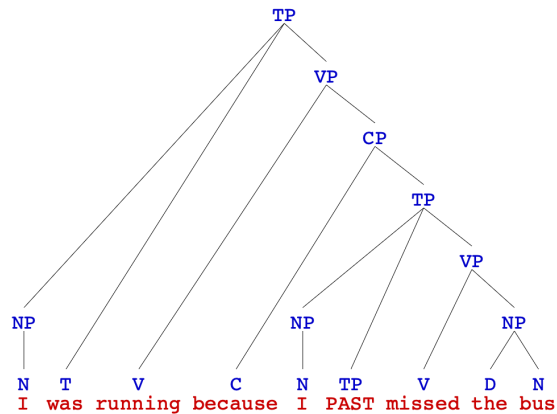
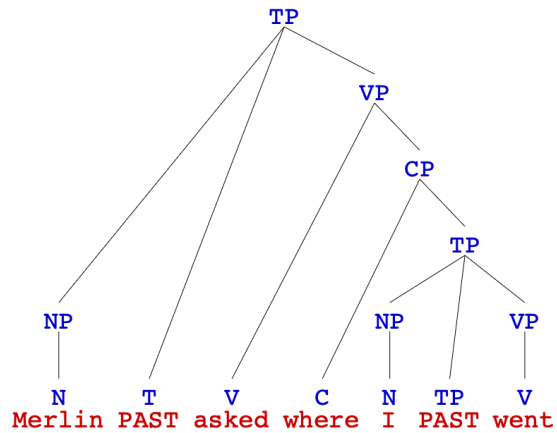
- dependent clauses are usually linked to main clauses with a special type of word called a **complementiser** or **subordinating conjunction** (relative pronouns are a subtype):
 - *that, which, because, if, so, since, which, who, although, where...*
 - *She said [_{CP} **that** she went to China] .*
 - *I bought an umbrella [_{CP} **because** it was raining] .*
 - *Please tell me [_{CP} **if** you're going to the store] .*
- sometimes the complementiser is **omitted** (often in English, that can be variably omitted):
 - *She said [_{CP} **(that)** she went to China] .*
 - *The book [_{CP} **(that)** I read] is about dinosaurs.*
 - *Ben thought [_{CP} **(that)** he was going to be late] .*

Structure of Dependent Clauses

- dependent clauses are **embedded sentences**, so we will also take them as a **TP**
- dependent clause **TPs** are then inserted into a larger constituent called the **complementizer phrase** or **CP**: **CP** → **(C) TP**
 - C (complementizer), is optional
- CP** is then inserted into the **main clause**
- note** that some dependent clauses don't have an explicit subject NP
 - He wants [TP Ø to go to Machu Picchu].*
 - I know the author [CP who [TP Ø wrote this book]].*



CPs inside VPs



- four roles **CP** can play in a sentence:

1. take the role of an **NP**, acting as the **object**

- Merlin [**VP** asked [**NP** a question]].
- Merlin [**VP** asked [**CP** where I went]].

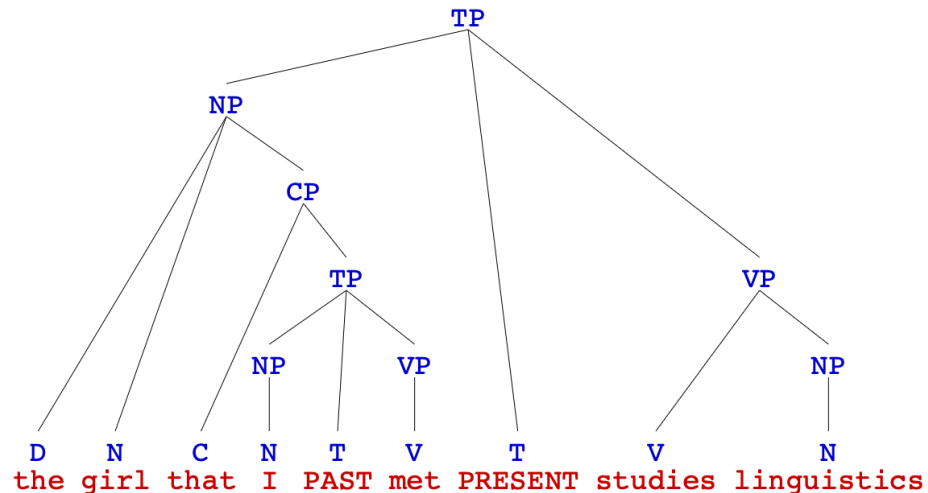
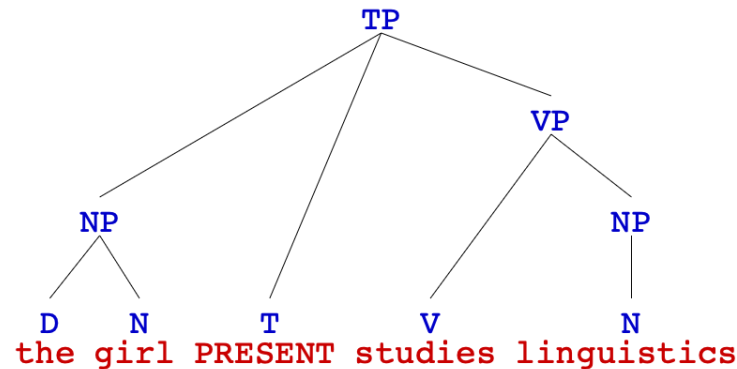
2. take the role of **AdvP**, acting as an **adjunct**

- I was [**VP** running [**AdvP** quickly]].
- I was [**VP** running [**CP** because I missed the bus]].

- since these **CP**s go inside **VP**s, let's modify our **VP** rule to account for these two cases:

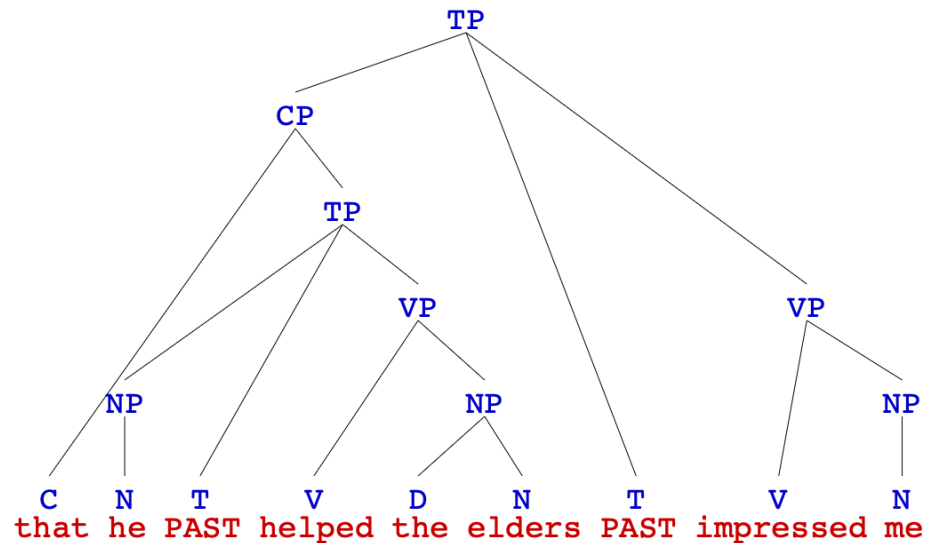
- VP** → (AdvP+) **V** (NP) ({**NP**, **CP**}) (AdvP+) (PP+) (AdvP+) (**CP**+) (AdvP+)

CPs inside NPs



- four roles **CP** can play in a sentence:
- act **like** an **AdjP**, modifying a noun and going inside an **NP**
 - [**NP** The girl] studies linguistics.
 - [**NP** The girl [**CP** that I met]] studies linguistics.
- these can go inside an **NP** anywhere in the sentence:
 - [**NP** The book [**CP** I read]] is about geology.
 - She bought [**NP** the book [**CP** I read]].
- since these **CP**s go inside **NPs**, let's modify our **NP** rule:
 - **NP** → (D) (AdjP+) **N** (PP+) (**CP**+)

CPs as the Subject



- four roles **CP** can play in a sentence:
- act as **the subject**, going into a **TP**
 - *[**TP** [**NP** That story] impressed me]*.
 - *[**TP** [**CP** That he helped the elders] impressed me]*.
- since these **CP**s go inside **TP**s, let's modify our **TP**s rule:
 - **TP**s → {**NP**, **CP**} (**T**) VP

Modified Phrase Structure Rules for English

- **NP** $\rightarrow$ (D) (AdjP+) **N** (PP+) (CP+)
- **TP** $\rightarrow$ {**NP**, **CP**} (**T**) VP
- **CP** $\rightarrow$ (**C**) TP
- **PP** $\rightarrow$ **P** NP
- **AdjP** $\rightarrow$ (AdvP) **Adj**
- **AdvP** $\rightarrow$ (AdvP) **Adv**
- **VP** $\rightarrow$ (AdvP+) **V** (NP) ({**NP**, **CP**}) (AdvP+) (PP+) (AdvP+) (CP+)

Drawing Complex Sentences

- if a sentence has **two main verbs** joined by a **complementizer**, it's **complex**
- this doesn't include sentences with two main verbs joined by a **coordinating conjunction** like and, or, but

*I read a book **and** I took a walk.*

- then identify which of these verbs is in the **main** clause, and which is in the **subordinate** clause
 - the **subordinate** clause is the one that **starts with a complementiser**
 - but be aware that the complementiser that is sometimes **omitted**
 - draw the tree for the subordinate clause first
- draw the tree for the **main** clause without the subordinate clause, then figure out **how the two are connected**
 - does the subordinate clause modify a noun (goes in to **NP**)
 - is it part of the main clause's VP (goes in to **VP**)
 - is the subject (goes into **TP**)

Exercise III

complex sentences

draw **syntax trees** for the following sentences.

- John said that he likes pizza.
- The dog I saw can run fast.
- I will take an umbrella if it rains.
- I saw a deer that was grazing in the park.
- I am studying because I am taking a test tomorrow.
- That Merlin randomly did his part of work totally ruined the whole plan..

NP → (D) (AdjP+) **N** (PP+) **(CP+)**

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

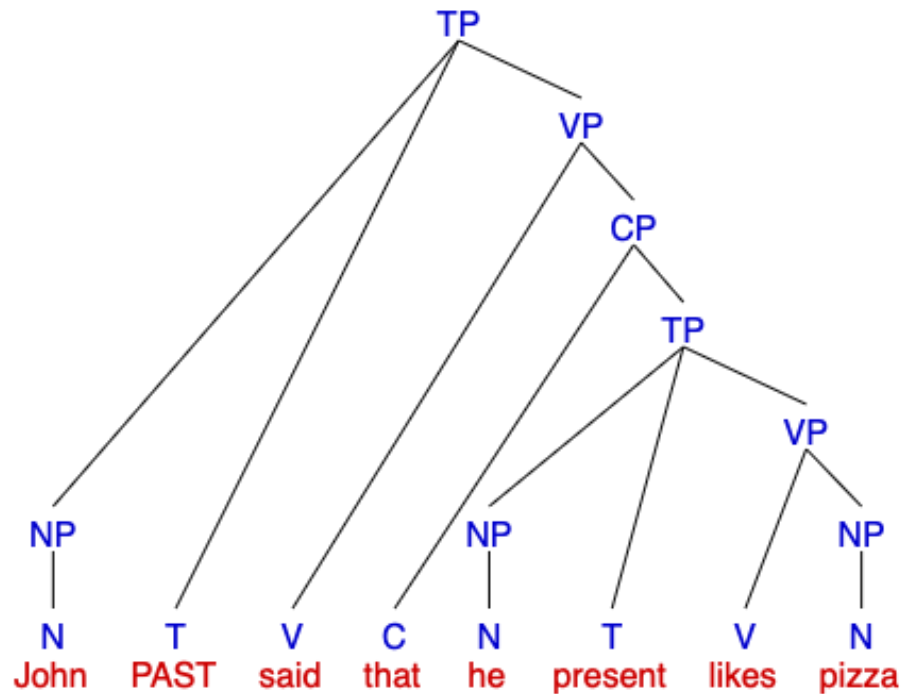
TP → {**NP, CP**} **(T)** VP

CP → **(C)** TP

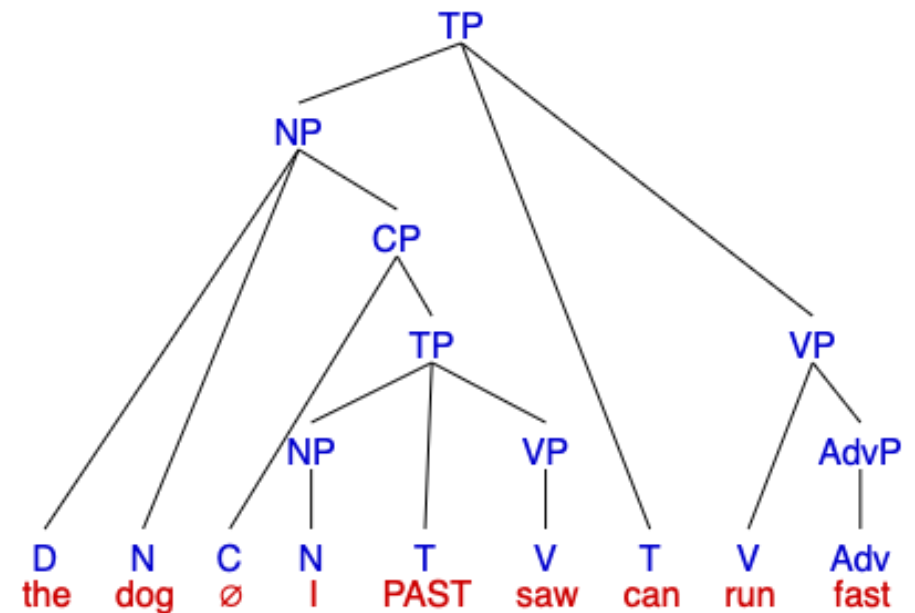
VP → (AdvP+) **V** (NP) (**{NP, CP}**)
(AdvP+) (PP+) (AdvP+) **(CP+)**

Exercise III

complex sentences



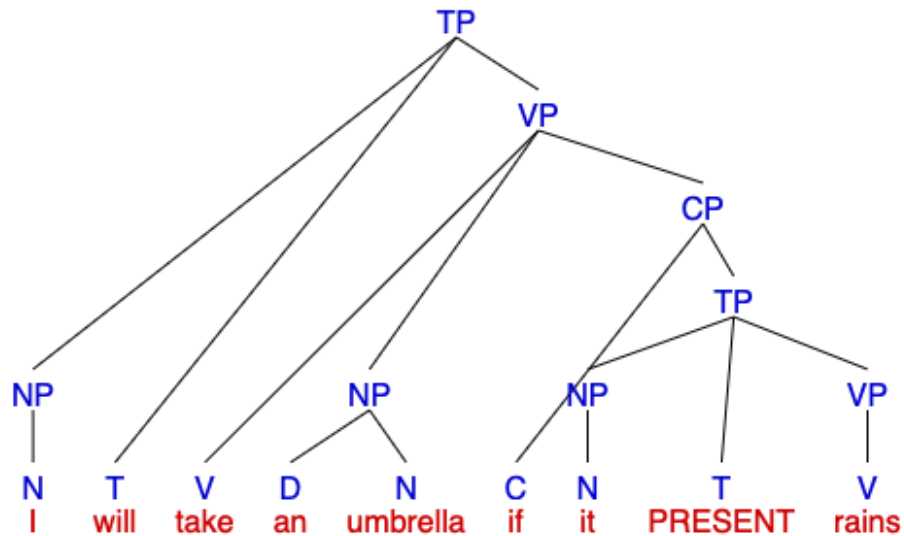
a. 'John said that he likes pizza.'



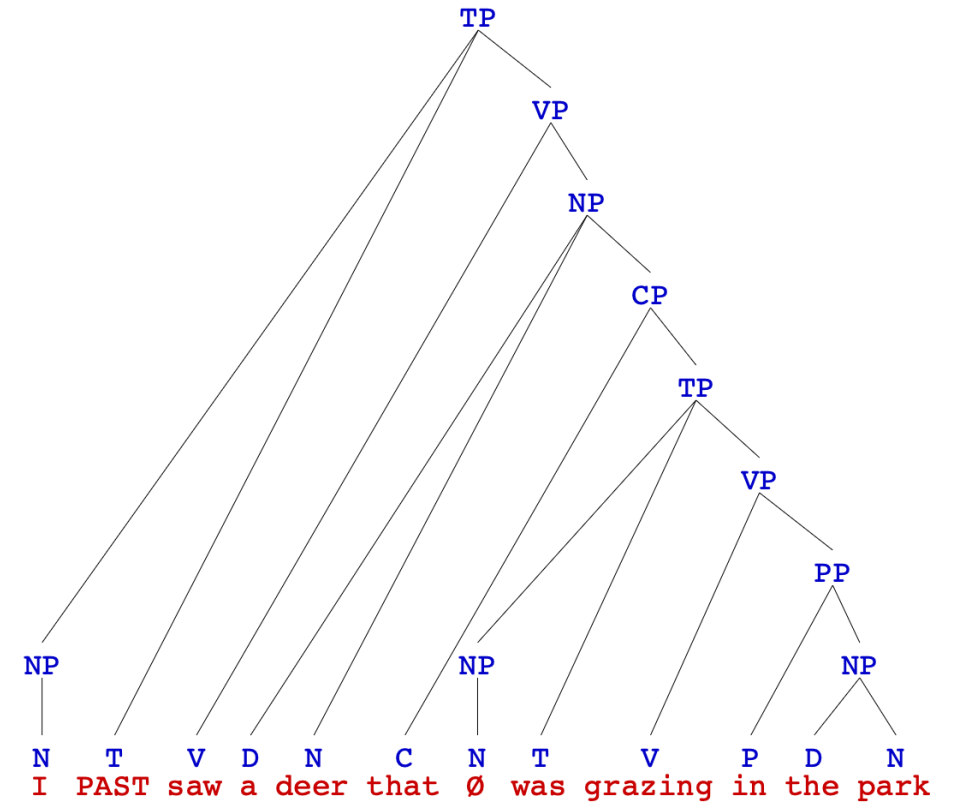
b. 'The dog I saw can run fast.'

Exercise III

complex sentences



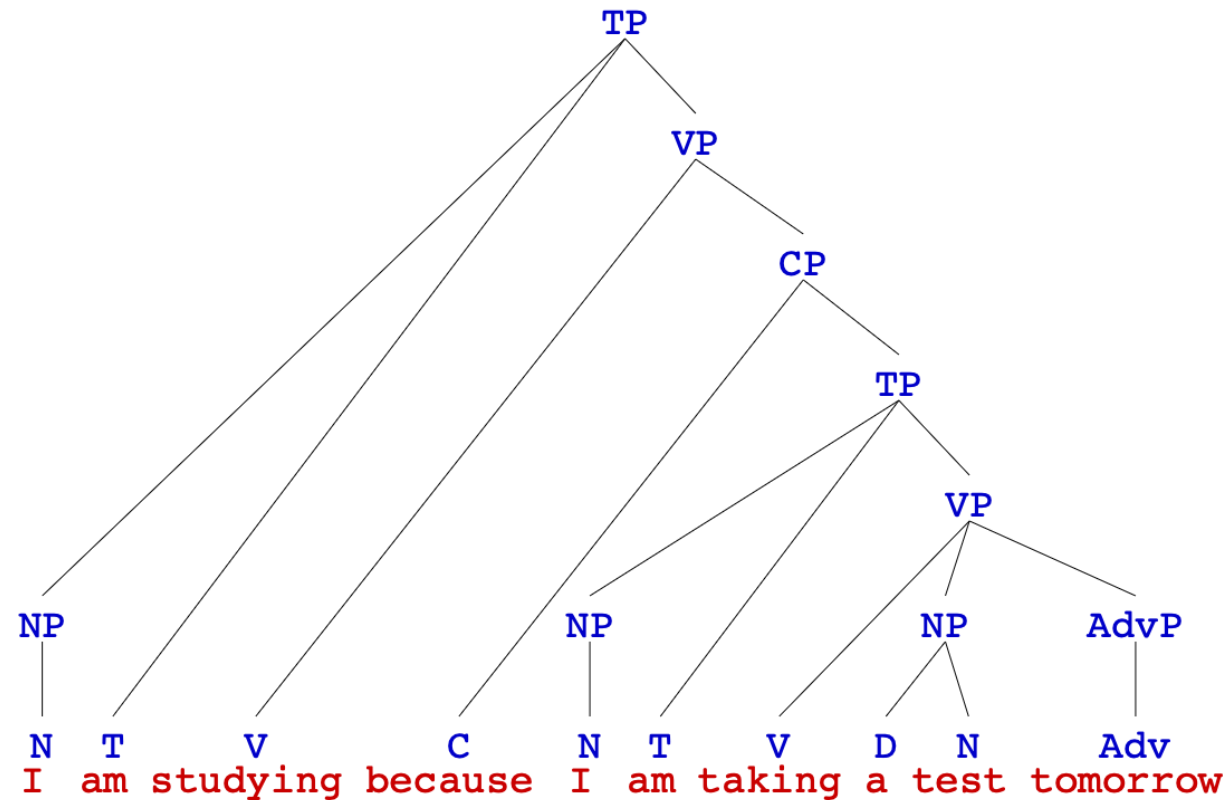
c. 'I will take an umbrella if it rains.'



d. 'I saw a deer that was grazing in the park.'

Exercise III

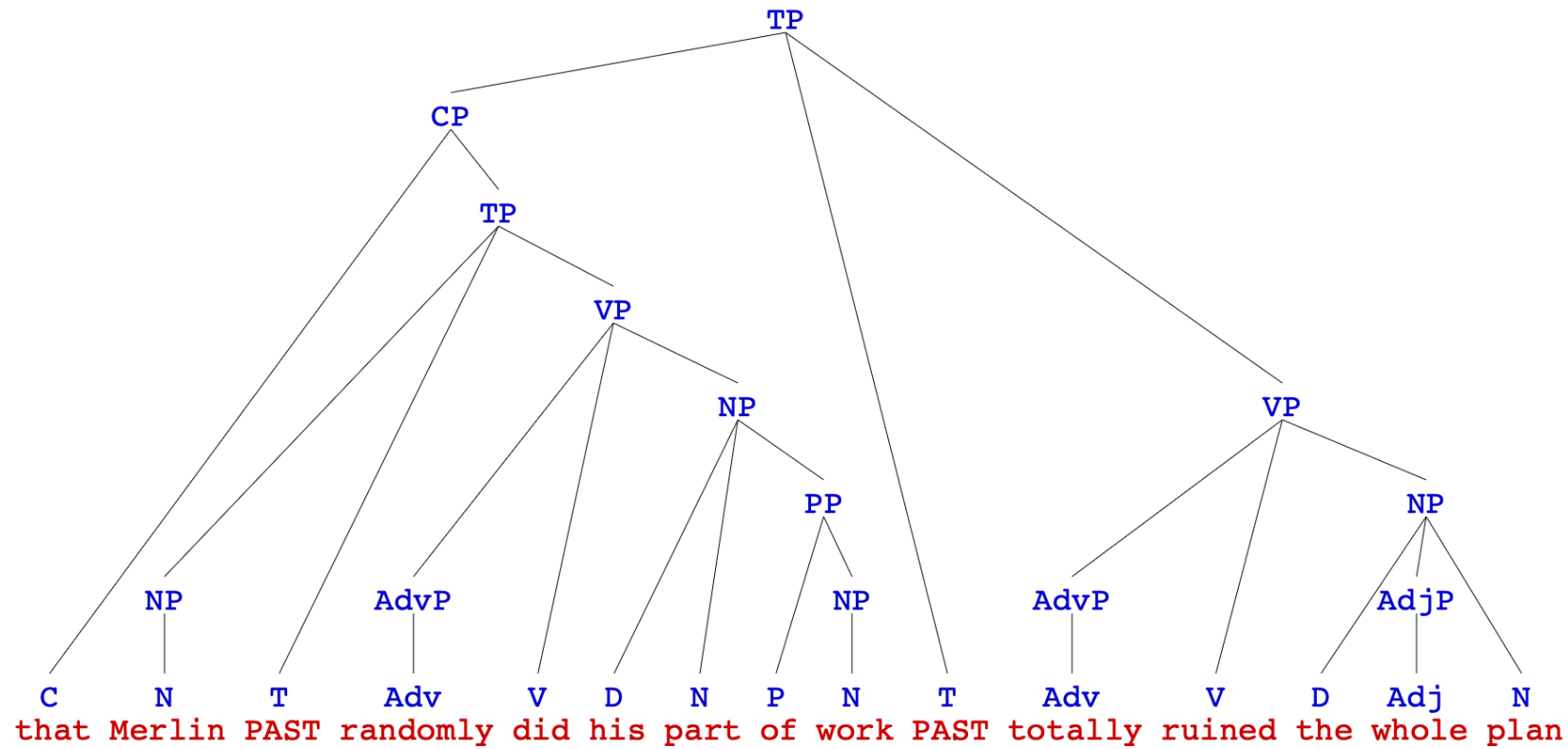
complex sentences



e. 'I am studying because I am taking a test tomorrow.'

Exercise III

complex sentences



f. 'That Merlin randomly did his part of work totally ruined the whole plan.']

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

Thai

(a) dèg khon níi kamlang kin
 boy CL this PROG eat
'This boy is eating.'

(b) mǎa tua nán kin khâaw
 dog CL that eat rice
'That dog ate rice.'

NP → (D) (AdjP+) **N** (PP+) (CP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → {**NP**, **CP**} (**T**) VP

CP → (**C**) TP

VP → (AdvP+) **V** (NP) ({**NP**, **CP**})
 (AdvP+) (PP+) (AdvP+) (**CP**+) (CP+)

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NP → (D) (AdjP+) **N** (PP+) (CP+)
PP → **P** NP
AdjP → (AdvP) **Adj**
AdvP → (AdvP) **Adv**
TP → {**NP**, **CP**} (**T**) VP
CP → (**C**) TP
VP → (AdvP+) **V** (NP) ({**NP**, **CP**})
 (AdvP+) (PP+) (AdvP+) (**CP**+)

Thai

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 'That dog ate rice.'

Thai vs English

1. Thai has **classifiers** (no English equivalent)
2. demonstratives (**this / that**) follow the noun in Thai but precede it in English
3. the **progressive** is expressed by a **separate** word in Thai and the verb does not change form

Cross-linguistic Comparison

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NP → (D) (AdjP+) **N** (PP+) (CP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → {**NP**, **CP**} (**T**) VP

CP → (**C**) TP

VP → (AdvP+) **V** (NP) ({**NP**, **CP**})
(AdvP+) (PP+) (AdvP+) (**CP**+) (CP+)

French

(a)	cet	homme	intelligent	comprendra	la	question
	this	man	intelligent	will understand	the	question

'This intelligent man will understand the question.'

(b)	ces	hommes	intelligents	comprendront	les	questions
	these	men	intelligent.PL	will understand.PL	the.PL	questions.PL

'These intelligent men will understand the questions.'

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

NP → (D) (AdjP+) **N** (PP+) (CP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → {**NP**, **CP**} (**T**) VP

CP → (**C**) TP

VP → (AdvP+) **V** (NP) ({**NP**, **CP**})
(AdvP+) (PP+) (AdvP+) (**CP**+) (CP+)

Greek

(a)	ego	edos-a		s-to	skyl-o		en-a		kokkal-o
	ego	edos-	a	s-	to	skyl-	o	en-	a
	I	give-	PST.1SG	to	the	dog-	OBJ.SG	one-	SG.NEUT
								bone-	SG.NEUT

I gave a bone to the dog.

OBJ = object marker

NEUT = neuter noun (in contrast with feminine and masculine)

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

Japanese

(a)	watashi	ga	sakana	o	tabete	iru
	I	SUBJ	fish	OBJ	eat.PROG	am
	<i>'I am eating fish.'</i>					

SUBJ = subject marker

OBJ = object marker

NP → (D) (AdjP+) N (PP+) (CP+)

PP → P NP

AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

TP → {NP, CP} (T) VP

CP → (C) TP

VP → (AdvP+) V (NP) ({NP, CP})
(AdvP+) (PP+) (AdvP+) (CP+)

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

NP → (D) (AdjP+) **N** (PP+) (CP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → {**NP**, **CP**} (**T**) VP

CP → (**C**) TP

VP → (AdvP+) **V** (NP) ({**NP**, **CP**})
(AdvP+) (PP+) (AdvP+) (**CP**+) (CP+)

Swahili

(a)	mtoto		alivunja		kikombe	
	m-	toto	a- li-	vunja	ki-	kombe
	NOUN CLASS	child	he PST	break	NOUN CLASS	cup
	'The child broke the cup.'					

(b)	watoto		wanavunja		vikombe	
	wa-	toto	wa- na-	vunja	vi-	kombe
	NOUN CLASS	child	they PRES	break	NOUN CLASS	cup
	'The children break the cups.'					

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

NP → (D) (AdjP+) N (PP+) (VP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → {NP, CP} **(T)** VP

CP → **(C)** TP

VP → (AdvP+) **V** (NP) ({NP, CP})
(AdvP+) (PP+) (AdvP+) (**CP+**)

Korean

(a)	ki	sonyɔn-iee		wiyu-lil		masi-ass-ta	
	ki	sonyɔn-	iee	wiyu-	lil	masi-	ass- ta
	the	boy	SUBJ	milk	OBJ	drink	PST ASSRT
	'The child broke the cup.'						

(b)	ki-nin	muɔs-ɪl		mɔk-ass-ninya	
	ki	nin	muɔs-	ɪl	mɔk- ass- ninya
	he	SUBJ	what	OBJ	eat past QUES
	'The children break the cups.'				

Cross-linguistic Comparison

state at least **three differences** between English and the following languages, using just the sentence(s) given, **ignore** lexical differences (i.e., the different vocabulary)

Tagalog

(a)	Nakita	ni	Pedro-ng		puno	na	ang	bus
	Nakita	ni	Pedro	-ng	puno	na	ang	bus
	saw	ART	Pedro	that	full	already	TOP	bus
	<i>'Pedro saw that the bus was already full.'</i>							

ART = article

TOP = topic marker

NP → (D) (AdjP+) N (PP+) (CP+)

PP → P NP

AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

TP → {NP, CP} (T) VP

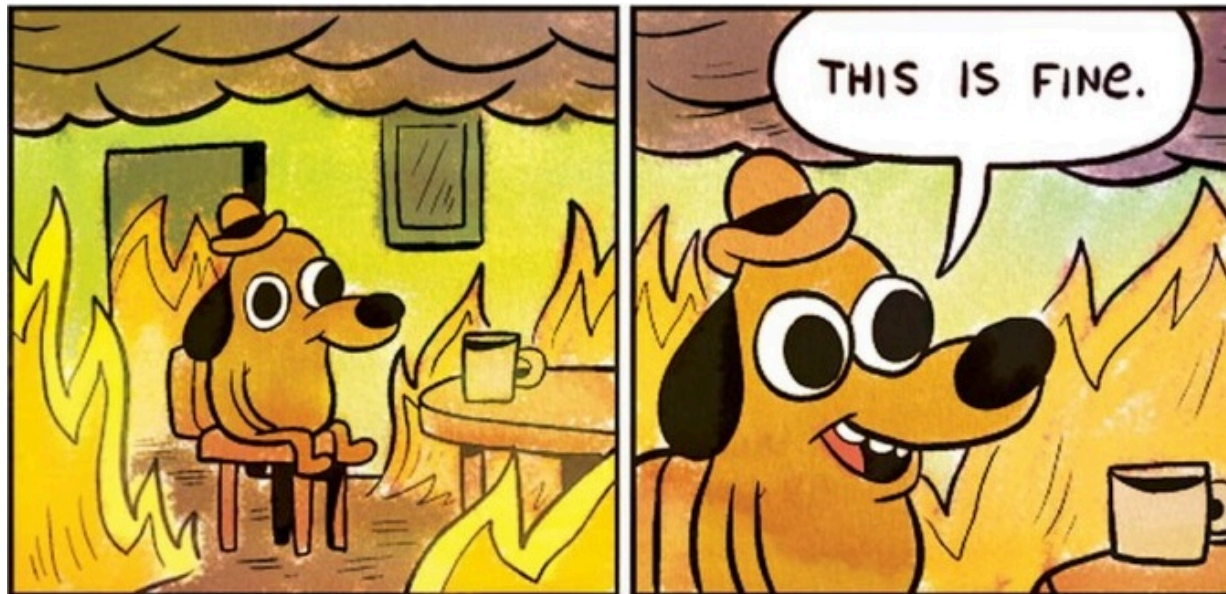
CP → (C) TP

VP → (AdvP+) V (NP) ({NP, CP})
(AdvP+) (PP+) (AdvP+) (CP+)

Assignment III is due this Sunday (**Oct 11**) at **11:59 pm**

start preparing for the **MIDTERM**

the **MIDTERM** is on **Oct 14** at class time



Slides created via the R package **xaringan**.